Employee Performance Data Analysis Report

1. Introduction

This report provides an analysis of the employee dataset to understand key performance indicators, salary distributions, and the relationship between various employee attributes such as experience, training, and satisfaction. The goal is to extract actionable insights that can help in optimizing workforce management and employee satisfaction.

2. Data Selection

The data set used for this analysis is employee.csv, which contains detailed information for 1,000 employees. The dataset includes the following attributes:

- **Employee Identification**: Employee_ID, Name
- **Demographics**: Age, Gender
- **Professional Details**: Department, Position, Years_of_Experience, Salary
- **Performance Metrics**: Performance_Score, Performance_Category, Projects_Completed, Attendance_Rate, Training_Hours
- **Employee Sentiment**: Satisfaction_Score, Satisfaction_Level
- **Career Growth**: Promotion_Eligible

Code:

```
import pandas as pd
emp=pd.read_csv('employee.csv')
emp.iloc[0:5,0:3]
```

Output:

	Employee_ID	Name	Age
0	EMP0001	Chen Rao	23
1	EMP0002	Lee Joshi	56
2	EMP0003	Priya Verma	35
3	EMP0004	Robert Jackson	48
4	EMP0005	Rahul Smith	49

3. **Key Libraries Used:**

- pandas: Used for data loading, cleaning, grouping, and calculating means/correlations.
- NumPy: Used for fast mathematical operations and calculating the promotion percentage.
- matplotlib: The foundational plotting library used to create the layout of the dashboard.
- Seaborn: A high-level visualization library used to make the charts (histograms, scatter plots, and count plots) look professional.

4. Initial Analysis

4.1 Data Cleaning

- The initial data inspection revealed:
- Header Cleaning
  Code :
  ```
  print(emp.columns.tolist())
  emp.columns=emp.columns.str.strip().str.lower().str.replace("_", "")
  print("Fixed Columns")
  print(emp.columns.tolist())
  ```
  Output:
  ['Employee_ID', 'Name', 'Age', 'Gender', 'Department', 'Position', 'Years_of_Experience', 'Salary', 'Performance_Score', 'Projects_Completed', 'Attendance_Rate', 'Training_Hours', 'Satisfaction_Score', 'Performance_Category', 'Satisfaction_Level', 'Promotion_Eligible']
  Fixed Columns
  ['employeeid', 'name', 'age', 'gender', 'department', 'position', 'yearsofexperience', 'salary', 'performancescore', 'projectscompleted', 'attendancerate', 'traininghours', 'satisfactionscore', 'performancecategory', 'satisfactionlevel', 'promotioneligible']

- **Duplicates**: No duplicate records were found in the dataset...

  ```
  #it is check for duplicates values if any
  ```
  Code:
  ```
  emp.duplicated().sum()
  ```
  Output: np.int64(0)

- **Drop Duplicates**: No duplicate records were found in the dataset...

- ```
  #it is check for drop duplicate  values if any
  ```
  Code:
  ```
  emp.drop_duplicates()
  ```

  output:
  ```

	employeeid	name	age
0	EMP0001	Chen Rao	23
1	EMP0002	Lee Joshi	56
2	EMP0003	Priya Verma	35

- **Null Values**: There were no missing values across any of the columns.

```
#it is check for duplicated values if any
```

Code:

```
emp.isna.sum()
```

Output:

	0
employeeid	0
name	0
age	0
gender	0
department	0

dtype: int64

5. Feature Selection

To enhance the analysis, the following features of engineering steps were performed:

- **Adding Columns**:

Code:

```
emp['SalaryperYearExp']=emp['salary']/emp['yearsofexperience']+1
emp['SalaryperYearExp'].head(5)
```

Output:

	SalaryperYearExp
0	38644.000000
1	5146.263158
2	12655.090909

	SalaryperYearExp
3	4938.892857
4	5316.814815

dtype: float64

- **Removing Column**:

  Code:
  ```
  emp.drop(columns='traininghours')
  print("column Dropped")
  ```
  Output: `column Dropped`

- **Updating Column**:

  Code:

  # Updating column (Example: mapping satisfaction level to numeric for correlation

  performance_mapping = {

  'Needs Improvement': 1,

  'Meets Expectations': 2,

  'Exceeds Expectations': 3,

  'Outstanding': 4

  }

  emp['performance_numeric'] = emp['performancecategory'].map(performance_mapping)

  emp['performance_numeric'].head()

Output:

	performance_numeric
0	3.0
1	4.0
2	1.0
3	1.0
4	2.0

6. Descriptive Statistics

The descriptive summary of the numerical features is as follows:

Code:

```
emp.describe()
```

Output:

Metric	Age	Years of Exp	Salary	Perf. Score	Projects	Training Hrs	Sat. Score
Mean	41.35	20.88	$147,639	69.40	29.97	38.96	4.61
Std Dev	11.02	11.25	$39,608	16.01	15.58	14.62	2.02
Min	22.00	0.00	$35,000	30.00	0.00	10.00	1.00
50% (Median)	41.00	21.00	$153,786	70.00	30.00	39.00	5.00
Max	60.00	43.00	$209,841	100.00	50.00	87.00	10.00

Code:

```
Information=emp.info()
```

Output:

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1000 entries, 0 to 999
Data columns (total 18 columns):
 #   Column          Non-Null Count  Dtype
---  ------          --------------  -----
 0   employeeid      1000 non-null   object
 1   name            1000 non-null   object
 2   age             1000 non-null   int64
 3   gender          1000 non-null   object
dtypes: float64(3), int64(7), object(8)
memory usage: 140.8+ KB
```

7. Analysis of Questions and Insights

7.1 Pandas Questions

1. **Which department has the highest average salary?**

Code:

```
import pandas as pd
high_avg_department=emp.groupby('department')['salary'].mean().idxmax()
high_avg_salary=emp.groupby('department')['salary'].mean().max()
print(high_avg_department)
print(high_avg_salary)
```

Output:

```
Legal
183947.84403669724
```

2. **What is the distribution of employees across different performance categories?**

Code:

```
#Q2:Distribution of employees across performance categories
performance_distribution=emp['performancecategory'].value_counts()
print(performance_distribution)
```

Output:

```
performancecategory
Meets Expectations      352
Exceeds Expectations    252
Needs Improvement       212
Outstanding             122
Unsatisfactory           62
Name: count, dtype: int64
```

3. **Who are the top 5 highest paid employees?**

Code:

```
# Q3: Top 5 highest paid employees
top5=emp.nlargest(5,'salary')[['name','salary']]
```

top5

Output:

	name	salary
247	**Aarav Reddy**	**209841**
15	**Olivia Reddy**	**209709**
317	**Park Robinson**	**209676**
203	**James Chen**	**209472**
220	**Choi Harris**	**209159**

4. **What is the correlation between** Years_of_Experience **and** Salary**?**

   Code:

   ```
   # Q4: Correlation between Years_of_Experience and Salary
   correlation=(emp['yearsofexperience']).corr(emp['salary'])
   correlation
   ```

   Output: np.float64(0.6949949369094003)

5. **Which gender has a higher average satisfaction score?**

   Code:

   ```
   # Q5: Which gender has a higher average satisfaction score?
   gender_satistaction_gender=emp.groupby('gender')['satisfactionscore'].mean()
   gender_satistaction_gender
   ```

Output:

	satisfactionscore
gender	
Female	**4.630219**

	satisfactionscore
gender	
Male	4.597586

dtype: float64

7.2 Numpy Questions

1. **What is the median salary across the company?**

   Code:

   ```
   import numpy as np
   median_salary=np.median(emp['salary'])
   median_salary
   ```

   Output: np.float64(153785.5)

2. **What is the standard deviation of Performance Score?**

   Code:

   ```
   # Q2:Standard deviation of Performance_Score
   std_salary=np.std(emp['performancescore'])
   std_salary
   ```

   Output: 16.006679824373354

3. **Calculate the percentage of employees who are eligible for promotion.**

   Code:

   ```
   # Q3: Percentage of employees eligible for promotion
   promo_eligible_pct = (np.sum(emp['promotioneligible'] == 'Yes') / len(emp)) * 100
   promo_eligible_pct
   ```

   Output: np.float64(36.0)

7.3 Visualization Questions

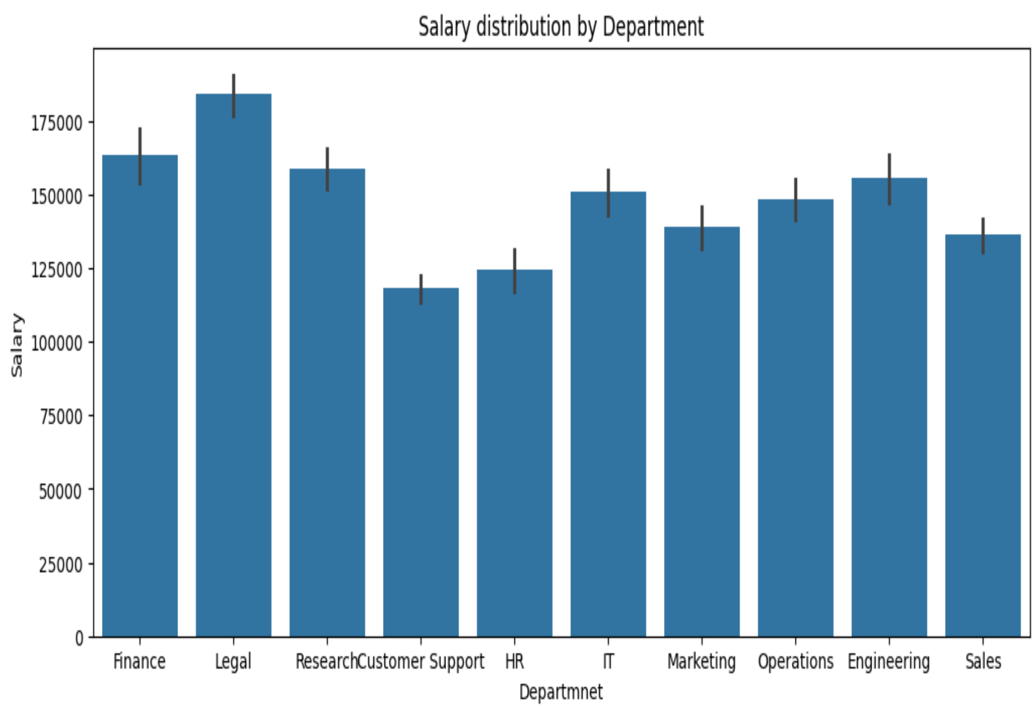
The following visualizations were generated to provide deeper insights:

1. **Salary Distribution by Department**: Highlights that Legal and other specialized departments maintain significantly higher pay scales than Customer Support or HR.

   Code:

```
#1: Salary distribution by Department
import matplotlib.pyplot as plt
import seaborn as sns
import numpy as np
plt.figure(figsize=(12,5))
sns.barplot(x='department',y='salary',
data=emp,estimator=np.mean,orient='vertical')
plt.title("Salary distribution by Department")
plt.xlabel("Departmnet")
plt.ylabel("Salary")
plt.show()
```

   Output:



2. **Age vs Salary**: Shows a general upward trend, though there is significant variance within the same age groups depending on the department.

Code:

```
#Shows a general upward trend, though there is significant variance within the same age groups depending on the department.

plt.figure(figsize=(10,6))

sns.scatterplot(x='age',y='salary',hue='department', data=emp)

plt.title("Avg Vs Salary")

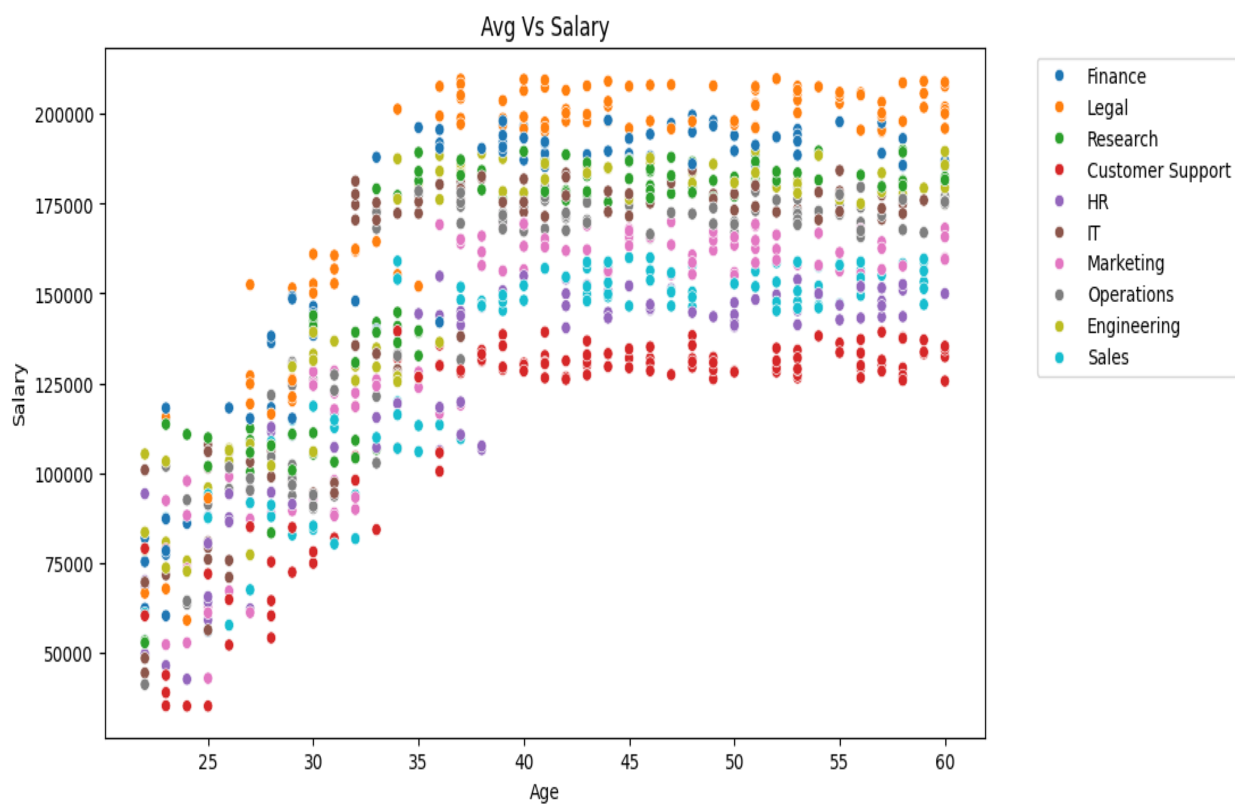
plt.xlabel("Age")

plt.ylabel("Salary")

plt.legend(bbox_to_anchor=(1.05, 1), loc='upper left')

plt.show()
```

Output:



3. **Performance Category Distribution** : A pie chart confirming that the majority of the workforce "Meets Expectations," while a small fraction is "Unsatisfactory."

Code:

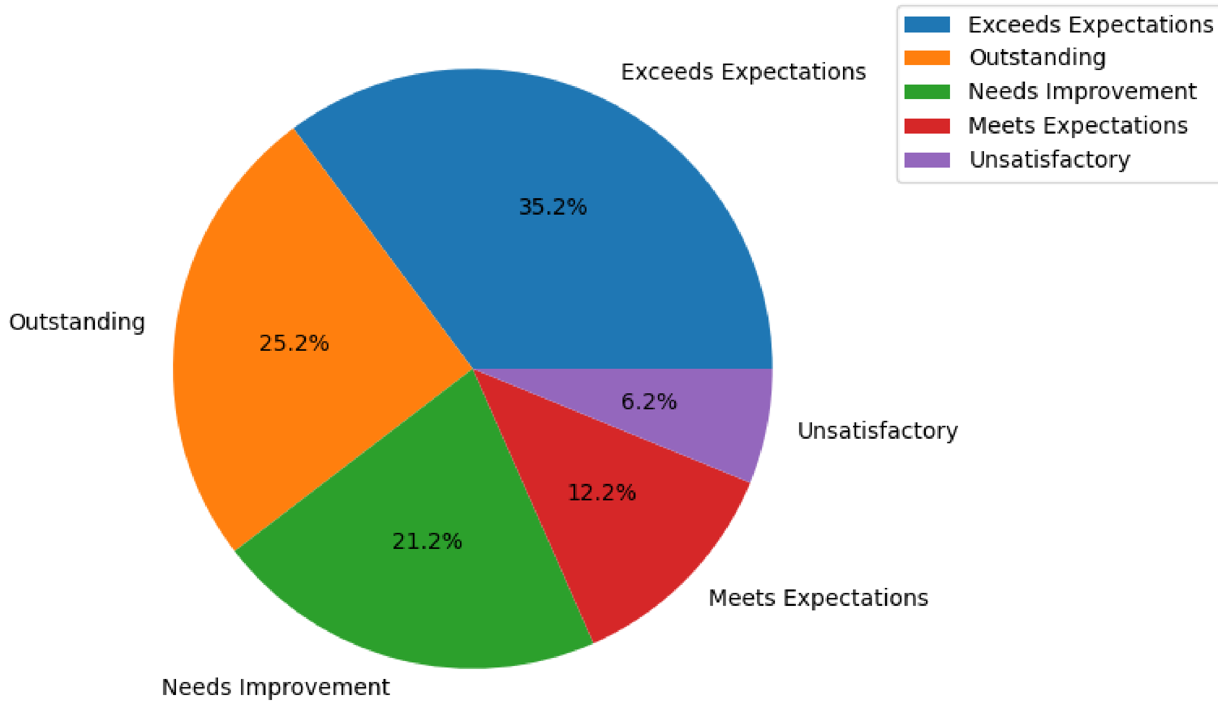
plt.figure(figsize=(10,6))

plt.pie(emp['performancecategory'].value_counts(), labels=emp['performancecategory'].unique(), autopct='%1.1f%%')

plt.legend(bbox_to_anchor=(1.05, 1), loc='upper left')

plt.show()

Output:

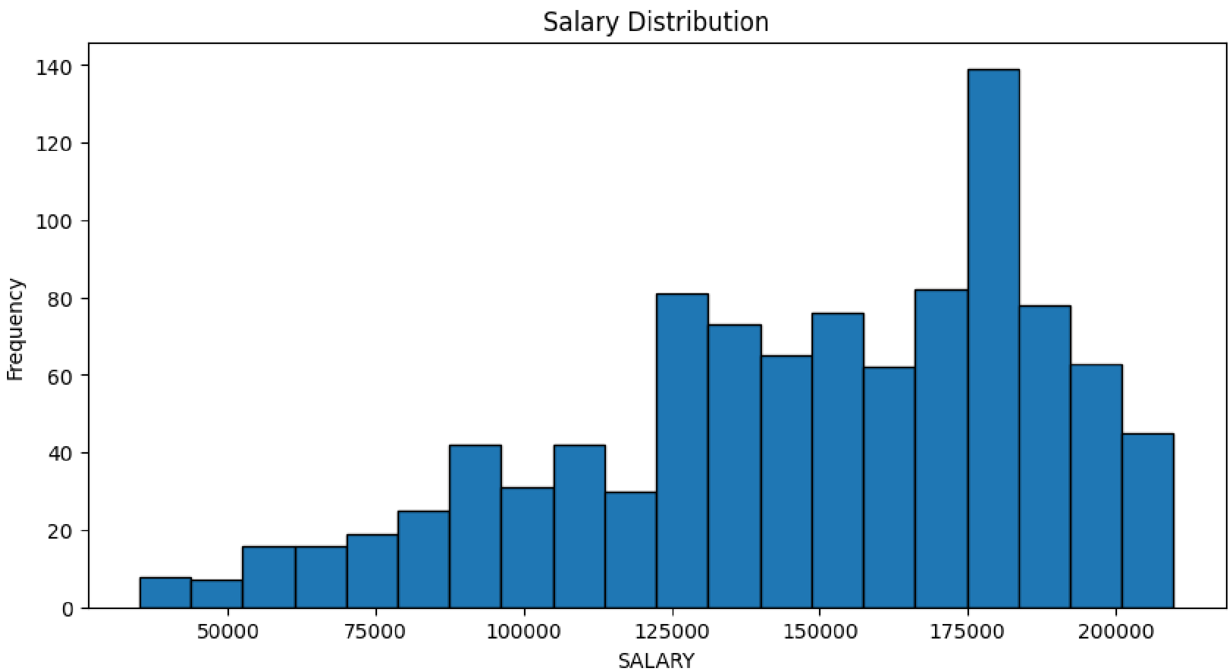


4. salary Distribution of Employees

Code:

```
import numpy as np
plt.figure(figsize=(10,5))
plt.hist(emp['salary'], bins=20, edgecolor='k')
plt.title("Salary Distribution")
plt.xlabel("SALARY")
plt.ylabel("Frequency")
plt.show()
```

Output:

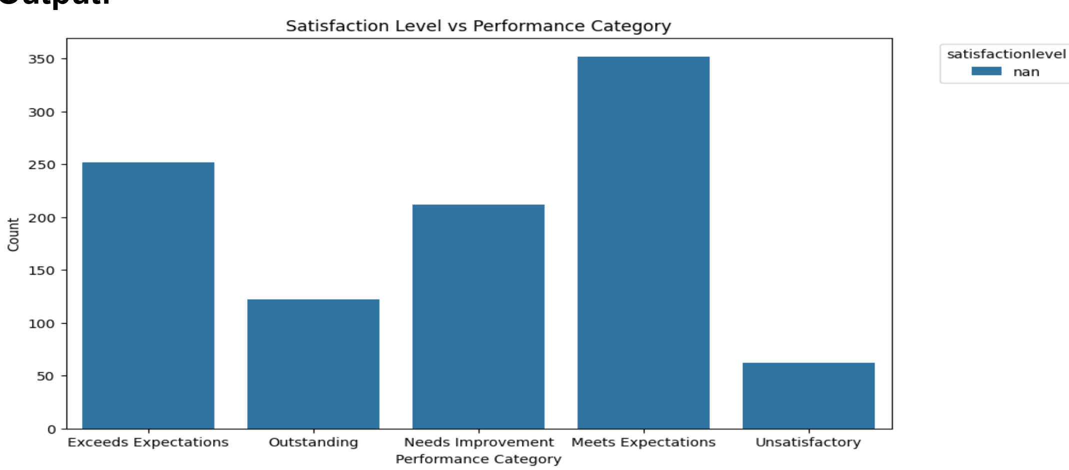


5. Count of Satisfaction Level vs Performance Category

Code:

```
plt.figure(figsize=(10,6))
sns.countplot(x='performancecategory', hue='satisfactionlevel', data=emp)
plt.xlabel("Performance Category")
plt.ylabel("Count")
plt.title("Satisfaction Level vs Performance Category")
plt.legend(title='satisfactionlevel' ,bbox_to_anchor=(1.05, 1), loc='upper left')
plt.show()
```

Output:



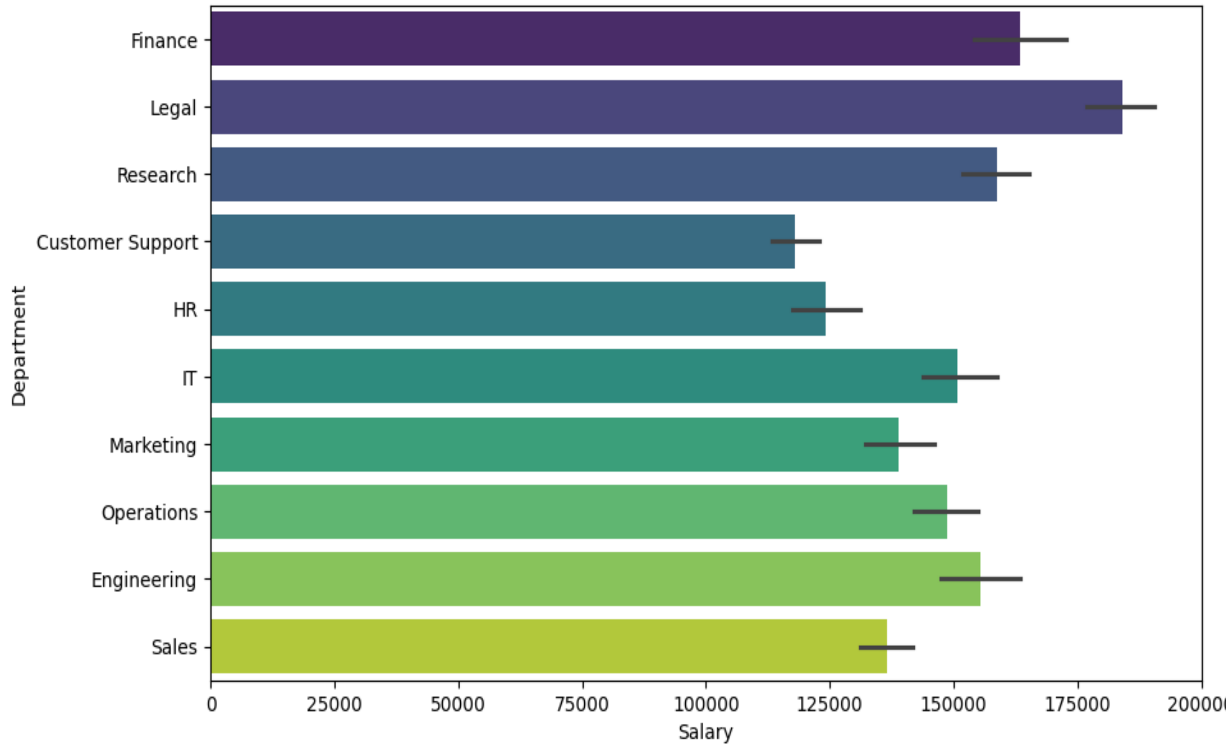
6. A horizontal bar chart highlights which departments are the highest and lowest paying.

Code:

```
plt.figure(figsize=(10,6))
```

```
sns.barplot(x='salary', y='department', data=emp, orient='h',hue='department'
,palette='viridis')
plt.xlabel("Salary")
plt.ylabel("Department")
plt.show()
```

Output:



Conclusion / Summary

The analysis reveals a healthy organization where salary is well-aligned with experience (correlation 0.69) and professional growth. The Legal department is the highest paying sector. While 36% of the staff are eligible for promotion, there is a clear link between continuous learning (Training Hours) and high performance. To further improve, the organization should focus on the "Needs Improvement" and "Unsatisfactory" groups (totaling ~27%) by analyzing their satisfaction levels and providing targeted training to shift them toward "Meets Expectations."